

Strong Exposure as a Sufficient Condition for Bidual Continuity

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2026-07-01

Source and Target

The source paper is S. Daptari, V. Montesinos, and T. S. S. R. K. Rao, *A study of weak*-weak points of continuity in the unit ball of dual spaces*, arXiv:2506.08458.

In Section 5.2, after a higher-dual preservation proposition, the authors write:

Related to Lemma 5.1, let us mention that we don't know of any geometric condition on $x \in S(X)$ in terms of X alone for ensuring that x should be a point of continuity of $id : (B(X^{**}), w^*) \rightarrow (B(X^{**}), w)$.

We shall resume the study of extreme points of continuity in Section 8 below.

5.2. Continuity in higher duals. Related to Lemma 5.1, let us mention that we don't know of any geometric condition on $x \in S(X)$ in terms of X alone for ensuring that x should be a point of continuity of $id : (B(X^{**}), w^*) \rightarrow (B(X^{**}), w)$.

We refer the reader to what was said at the beginning of Section 3 above regarding the bad stability behaviour—out of some classes of Banach spaces, see Section 6 for some instances—of points of w^* - w -continuity when going to higher duals and viceversa “coming down” and the appropriate examples

Result

We give a direct sufficient condition stated entirely in the original space. The conclusion is stronger than requested: it gives weak-star-to-norm continuity.

Theorem 1. *Let X be a real or complex Banach space. Suppose that $x \in S_X$ is a strongly exposed point of B_X , strongly exposed by $x^* \in S_{X^*}$, so that $\operatorname{Re} x^*(x) = 1$ and, for every $\varepsilon > 0$, there is $\delta > 0$ such that*

$$y \in B_X, \quad \operatorname{Re} x^*(y) > 1 - \delta \quad \implies \quad \|y - x\| < \varepsilon.$$

*Then the canonical image of x in X^{**} is a point of continuity of*

$$id : (B_{X^{**}}, \sigma(X^{**}, X^*)) \longrightarrow (B_{X^{**}}, \|\cdot\|).$$

*In particular, x is a point of continuity of $id : (B_{X^{**}}, w^*) \rightarrow (B_{X^{**}}, w)$.*

Proof Intuition

Strong exposure says that one weak slice cut out by x^* gives arbitrarily small norm neighborhoods of x inside the original unit ball. Goldstine's theorem says that B_X is weak-star dense in $B_{X^{**}}$. Therefore a bidual point lying in the same weak-star slice is a weak-star limit of original-space points eventually lying in the small norm slice. The norm ball around x is weak-star closed in the bidual, so the bidual point lies in the same small norm ball.

Proof

Fix $\varepsilon > 0$. By strong exposure, choose $\delta > 0$ such that

$$y \in B_X, \quad \operatorname{Re} x^*(y) > 1 - \delta \quad \implies \quad \|y - x\| < \varepsilon.$$

We claim that the weak-star slice

$$U_\delta = \{u \in B_{X^{**}} : \operatorname{Re} u(x^*) > 1 - \delta\}$$

is contained in $x + \varepsilon B_{X^{**}}$.

Let $u \in U_\delta$. By Goldstine's theorem, there is a net (y_α) in B_X which converges to u in $\sigma(X^{**}, X^*)$. Since $\operatorname{Re} u(x^*) > 1 - \delta$, eventually $\operatorname{Re} x^*(y_\alpha) > 1 - \delta$. Hence eventually $\|y_\alpha - x\| < \varepsilon$, i.e. $y_\alpha \in x + \varepsilon B_X \subset x + \varepsilon B_{X^{**}}$. The set $x + \varepsilon B_{X^{**}}$ is weak-star closed, because $B_{X^{**}}$ is weak-star compact and translations are weak-star homeomorphisms. Therefore $u \in x + \varepsilon B_{X^{**}}$.

Thus every norm neighborhood of x in $B_{X^{**}}$ contains a weak-star neighborhood of x relative to $B_{X^{**}}$. This is exactly weak-star-to-norm continuity at x . Since the weak topology on X^{**} is weaker than the norm topology, the asserted weak-star-to-weak continuity follows.

Verification and Limitations

The proof is purely functional analytic and uses only the definition of strong exposure and Goldstine's theorem. It gives a sufficient condition, not a characterization. It also says nothing about the later question in the source paper concerning stability of the dense G_δ property under subspaces, quotients, or c_0 -sums.

Run-level novelty checks were bounded: the lightweight indexes were searched for arXiv:2506.08458 and the main phrases “strongly exposed”, “bidual”, “points of continuity”, and “weak-star weak”. No prior run packet for this result was found. A local source search suggests the underlying lift of strong exposure to the bidual is standard, so the recommendation is to treat this as a useful identified sufficient condition rather than as a high-novelty theorem.

References

- S. Daptari, V. Montesinos, and T. S. S. R. K. Rao, *A study of weak*-weak points of continuity in the unit ball of dual spaces*, arXiv:2506.08458.
- J. Diestel, *Geometry of Banach Spaces: Selected Topics*, Lecture Notes in Mathematics 485, Springer, 1975. Background on denting and exposed points.